

Cainiao is Fast — But Not Everywhere

When does Alibaba's logistics network stop delivering its speed advantage?

Group 12

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The question

- **Audience: Cainiao** (Alibaba's logistics platform).
- Delivery speed drives sales and loyalty: online sales rise $\sim 1.45\%$ per business-day faster delivery (Fisher et al., 2019); late delivery erodes loyal customers (Rao et al., 2011).
- Cainiao handles **30.2%** of orders — does it actually deliver faster, and *everywhere*?

Data: 1.29M Tmall orders, Jan–Jul 2017 (order + 969 MB logistics + merchant files). Analysis in R; all visualizations built in Tableau.

Method: dig hierarchically, not in parallel

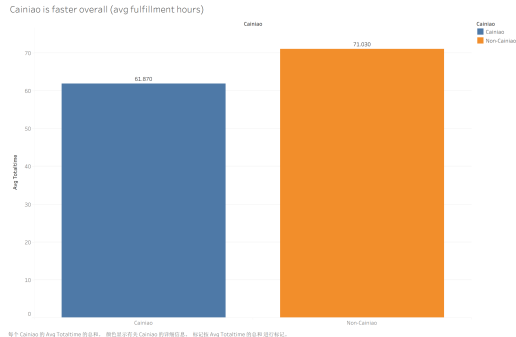
$$\text{totaltime} = \text{SIGNED} - \text{pay_timestamp (hours)}$$

- ① Cainiao vs. non-Cainiao
- ② + carrier size (Large >200k orders)
- ③ + destination-city size (top-2 signing cities = Big)
- ④ segment decomposition: pre-delivery / line-haul / last-mile

One focused theme; each level added only when the previous result motivates it.

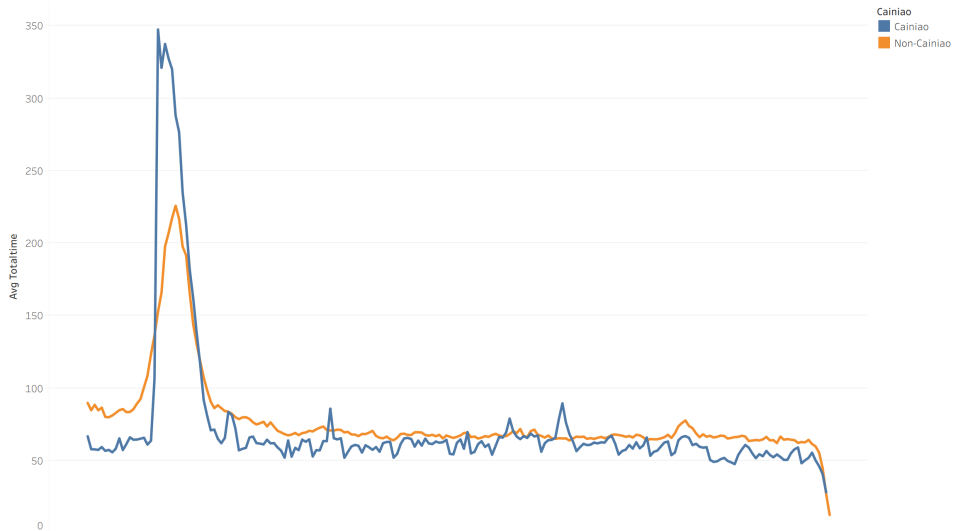
Headline: Cainiao is faster overall

- **61.9 h** vs. **71.0 h** $\Rightarrow$ ~ 9.2 h faster.
- Survives a fair control: among no-promise orders, 62.5 vs. 71.0 h.
- Stable across the whole 7-month window.



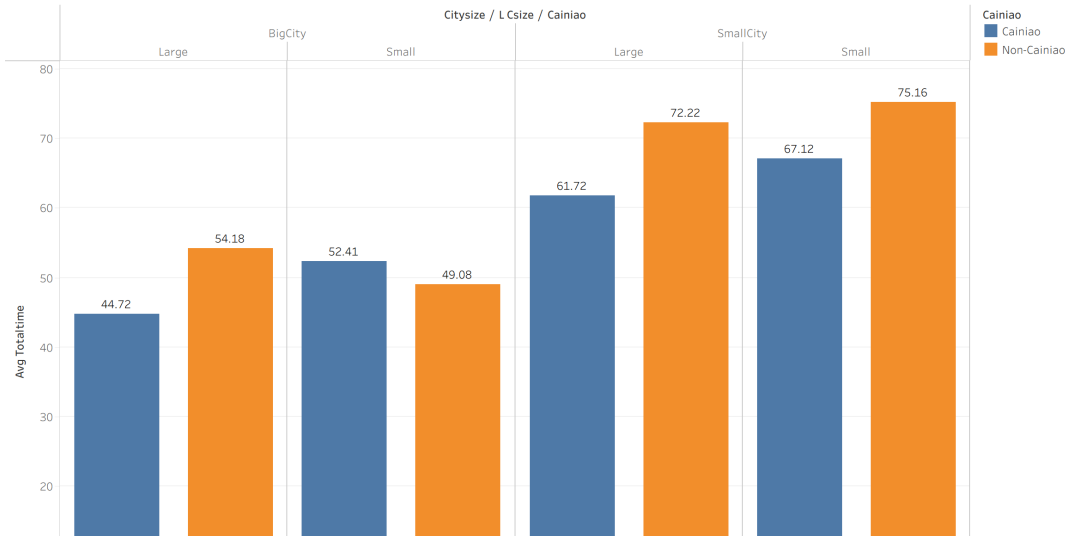
Stable over time

Daily average fulfillment time: Cainiao stays faster all year



...BUT the advantage is uneven — the punchline

Cainiao is faster everywhere — EXCEPT big-city destinations on small carriers (reversal)



The whole story on one screen

Cainiao is Fast - But Not Everywhere

1.29M

orders analysed

30.2%

handled by Cainiao

+9.2 h

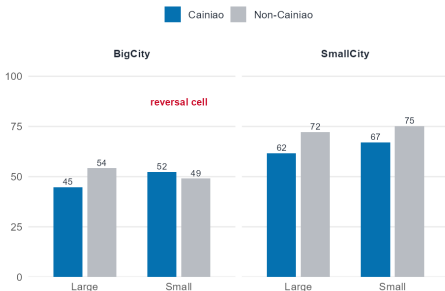
Cainiao faster overall

-3.3 h

Cainiao SLOWER:
big city x small carrier

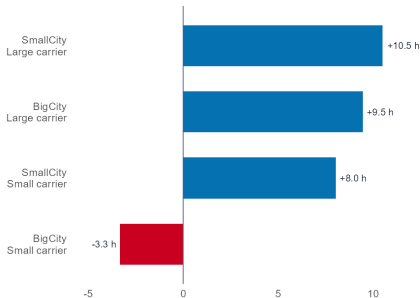
Cainiao is faster everywhere...

Avg fulfillment time (hours), lower is better



...but the advantage REVERSES in one cell

Hours Cainiao saves vs non-Cainiao (negative = slower)



Cainiao saves 8-10 hours in 3 of 4 cells - but for big-city destinations on small carriers it is 3.3 h slower, with the lowest reviews.

KPI strip + 2x2 breakdown + hours-saved (reversal -3.3 h in red).

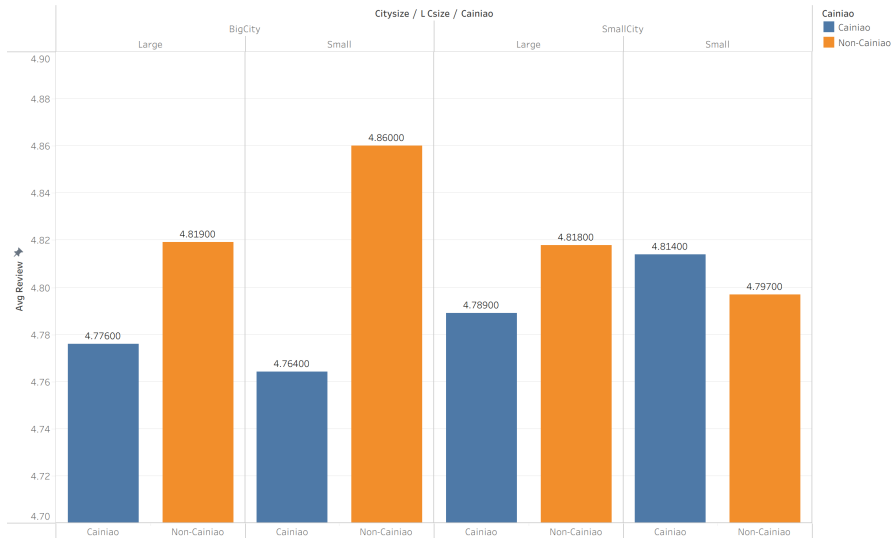
The reversal cell, in numbers

Destination	Carrier	Cainiao	Non-Cainiao	Cainiao saved
Big city	Large	44.7	54.2	+9.5 h
Big city	Small	52.4	49.1	−3.3 h
Small city	Large	61.7	72.2	+10.5 h
Small city	Small	67.1	75.2	+8.0 h

22,756 orders in the reversal cell — not a small-sample fluke.

Customers feel it too

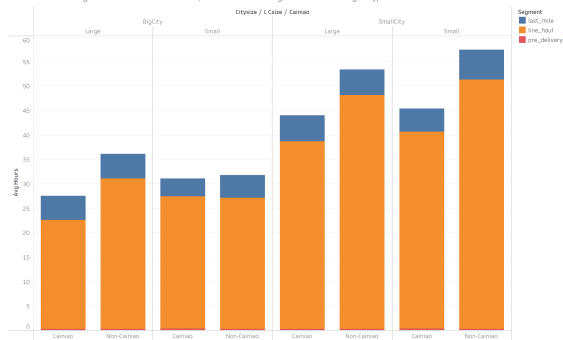
Customer reviews mirror the punchline: lowest for Cainiao in big-city/small-carrier



Mechanism: it's line-haul, not the last mile

- Last-mile is small & similar everywhere ($\sim 4\text{--}6\text{ h}$).
- Cainiao's edge = faster **line-haul** (e.g. 38.4 vs. 47.8 h).
- In the reversal cell that edge vanishes (27.0 vs. 26.9 h).

Where the time goes: line-haul dominates, and Cainiao's edge vanishes in big-city/small-carrier

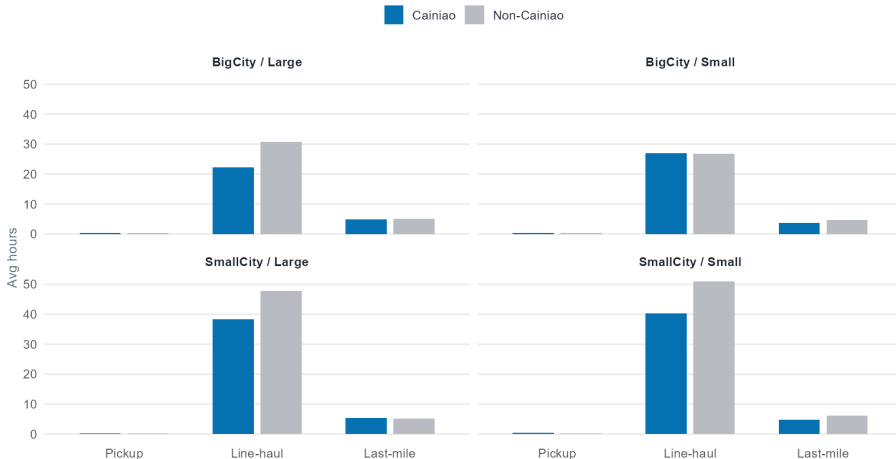


每个 Cainiao 的按 CitySize 与 L Csize 细分的 Avg Hours 的总称。颜色显示有关 Segment 的详细信息。

Mechanism, faceted: last-mile is flat, line-haul carries the gap

Where the time goes: it's line-haul, not the last mile

Avg hours per segment. Cainiao's line-haul edge vanishes for big-city / small-carrier orders.



An honest non-result

- Hypothesis: small carriers make more stops $\Rightarrow$ slower.
- **Not supported.** Small carriers pass through *fewer* cities (2.9–3.1) than large ones (3.4–3.7).
- So the penalty is *per-leg* transit/dwell time, not more hops — which sharpens the explanation we give.

- Big cities concentrate trunk-line volume; large carriers run dedicated, high-frequency line-haul, small carriers wait to consolidate (Yu et al., 2016).
- Small carriers can't execute one-size-fits-all smart routing; ignoring carrier heterogeneity degrades routing performance (Koç et al., 2016).

So what — three moves for Cainiao

- ① **Deepen ties with large carriers** on big-city lanes (watch concentration risk).
- ② **Make smart-routing carrier-capability-aware** (size, coverage, experience).
- ③ **Pilot big-city consolidation / parcel lockers** for small-carrier shipments to cut the line-haul + pickup penalty (Deutsch & Golany, 2018).

The weakness is concentrated, so the fix is cheap vs. a platform-wide change.

Summary in two sentences

Overall

Cainiao significantly shortens fulfillment time across the marketplace (~ 9 h, robust to a promise control).

Punchline

But that advantage largely disappears — and reverses — for big-city destinations served by small carriers, with the bottleneck in line-haul rather than the last mile.

Limitations: observational data (association, not causation); missing milestones; coarse big-city proxy.

Thank you — Group 12

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Questions?

Code, report & data pipeline:

<https://github.com/ericxuzhesheng/Buad345-final-project>